01-BOM

LOT Pager — Components Buying List (BOM v0.1)

Prices are **planning estimates, USD, qty-100 pricing, unverified** (vendor sites unreachable from build sandbox). Links are search/product-family URLs; confirm exact MPN and stock. Machine-readable copy: `bom.csv`.

Ref	Function	Part (MPN)	Why	Est. $/unit	Link
U1	MCU + Wi-Fi/BLE + camera DVP	Espressif ESP32-S3-MINI-1U-N8R2 *(or bare ESP32-S3R8 on custom PCB for v1)*	Only mainstream MCU with camera + Wi-Fi + low-power	4.0	https://www.espressif.com/en/products/socs/esp32-s3
U2	Environment (T/RH/P/gas, AI)	Bosch BME688	Weather + BSEC AI gas classification, 3×3×0.93 mm	9.0	https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme688/
U3	IMU w/ on-chip ML	ST LSM6DSOX	Machine Learning Core: tap/orientation/gesture without MCU	3.5	https://www.st.com/en/mems-and-sensors/lsm6dsox.html
U4	Ambient light	Vishay VEML7700	Auto-dim screen	1.5	https://www.vishay.com/en/product/84286/
CAM	2 MP camera	OmniVision OV2640 or GalaxyCore GC2145 FPC module (≤ 3.5 mm Z)	Cheap, ESP32 drivers exist	4.5	https://www.omnivision.com/products/
DSP	Screen	1.14” IPS 240×135, ST7789, SPI (≤1.6 mm panel)	Readable text, cheap	3.5	https://www.digikey.com/en/products/result?keywords=1.14%22%20ST7789
WPC	Qi RX coil + ferrite	20 mm Qi receiver coil, 0.35 mm flex + ferrite sheet	Thin	2.5	https://www.digikey.com/en/products/result?keywords=wireless%20charging%20receiver%20coil
U5	Qi RX IC	TI BQ51003 *(or BQ51013B)*	5 V Qi receiver, Li-ion path	3.0	https://www.ti.com/product/BQ51013B
U6	Battery charger/PMIC	TI BQ25185	Linear charger, power-path	1.5	https://www.ti.com/product/BQ25185
BAT	LiPo	301520 (≈ 3.0×15×20 mm, ~70 mAh) v0; 401230 alt	Thin; protected cell with PCM	3.0	https://www.digikey.com/en/products/result?keywords=lipo%20301520
HAP	Haptic (“pager buzz”)	6 mm coin LRA + DRV2605L driver	Silent pager feel	3.0	https://www.ti.com/product/DRV2605L
SW1	COPY button	3×3 mm low-profile tact (≤ 0.6 mm stroke) or capacitive-metal press	One button	0.4	https://www.digikey.com/en/products/result?keywords=low%20profile%20tact%20switch
LED	Camera-active LED (privacy)	0402 red	Trust	0.05	—
PCB	4-layer rigid-flex, 0.8 mm	PCBWay rigid-flex + PCBA	Fits stack	14.0	https://www.pcbway.com/rigid-flex-pcb.html
ANT	Antenna	ESP32 module PCB ant. via seam slot (v0) or 2.4 GHz ceramic chip	Needs non-metal gap	0.3	—
TOP	Front shell	316L stainless, CNC, bead-blast/satin		45–60	https://www.pcbway.com/cnc-machining-quote.html
BOT	Back shell	316L stainless, CNC, **mirror polish**		55–75	same
GLS	Window	0.5 mm chemically-strengthened glass / sapphire, AR	Display + camera + Qi path	6–10	https://www.pcbway.com/
GSK	Seam gasket	PEEK or epoxy band 0.8 mm	RF slot + seal	1.0	—
MISC	Screws M1.2 ×4, adhesive, ferrite, FPC			2.0	—

Charger (CHG): v0 = any off-the-shelf 5 W Qi pad (~$8) for dev. v1 = custom TX (e.g. TI BQ500212A or Renesas P9242) in a matching stainless disc with a recessed 40×40 pocket.

Estimated per-unit @100: ≈ $165–235. **Run total:** ≈ $16.5–23.5k + NRE.